

6/4/26

Day - 04

TCP :: 3-way handshake

- why does TCP works like this?

① TCP is Connection oriented

- unlike UDP who just fires packets & hope they arrive, TCP guarantees delivery. But first both sides must agree to talk.

2] Synchronize Sequence Nos -

- Each byte gets a sequence Number. The handshake lets both sides share their starting Number (ISN) which can help them track & detect missing values.

3] Negotiate parameter.

- window size, MSS (Max Segment Size) & options like SACK are agreed upon during handshake - before any real data flows.

States during 3-way handshake (SYN, SYN+ACK, ACK)

1] SYN -

From: client

Seq: 1000 (ISN)

Ack: None

Flags: SYN

Meaning: "My ISN is 1000".

2] SYN → ACK -

From: Server

seq 5000 (ISN)

Ack: 1001 (ISN+1)

Flags: SYN, ACK

Meaning: "Got 1000, my ISN is 5000".

3] ACK -

From: client

Seq: 1001

Ack: 5001 (ISN+1)

Flags: ACK

Meaning: "Got 5000, connection Ready!"

#3] TCP Flags -

- 1] SYN - Initiates Connection. Sets ISN only
first packet of a connection has
SYN = 1 without ACK = 1
- 2] ACK - Confirms receipt. ACK number = last
(Acknowledgement) received seq + 1. Almost
always set after the first SYN.
- 3] FIN - Careful connection teardown. uses
(finish) a 4-way process (FIN - ACK - FIN
- ACK)
- 4] RST - Abrupt termination. Seen in port
(Reset) scans, refused connection & network
errors
- 5] PSH - Tell receiver to pass data to
(Push) application immediately. don't buffer
Seen with HTTP request
- 6] URG - Data should be prioritized.